Lecture 14 - Natural proofs and everything else

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Remembering everything we've done in this course
The natural proofs barrier
- What is a natural proof?
- What else? Things we have not seen

The plan is to recap everything we have learned, to learn the natural proofs barrier of Razborov and Rudich, and to overview everything we haven't taught, the rest of the field of computational complexity.

Remembering everything we've done in this course

So far we have learned:

- How does one formalize a computational problem
- How does one formalize computation
- How does one measure the time required by a computation
- P, E, EXP, The time hierarchy theorem
- Non-determinism
- NP-hardness and NP-completeness
- The Cook-Levin theorem - SAT is NP-complete
- The web of reductions
- How to measure the space used by a computation
- PSPACE, NPSPACE, L, NL, the space hierarchy theorem, Savitch's theorem
- The relativization barrier
- Combinatorial models of computation: Formulas and circuits
- Upper-bounds for every function, and lower-bounds for a random function
- Cubic formula lower bounds
- Randomized computation; BPP, ZPP, RP, coRP
- Cryptography: Private-key crypto, one way functions, pseudorandom generators, the Goldreich-Levin theorem
- Derandomization - Hardness vs Randomness

Today:

The natural proofs barrier

The natural proofs barrier is a kind of independence result.

An equivalent way of stating $\mathsf{P} \neq \mathsf{NP}$ is saying that there is no polynomial-time algorithm for finding proofs of true statements, i.e., finding proofs is hard.

This may give rise to the following weird, somewhat creepy thought:

> What if $\mathsf{P} \neq \mathsf{NP}$ is true, and one of the proofs which is hard to find is the proof that $\mathsf{P} \neq \mathsf{NP}$?
>
> I.e., what if $\mathsf{P} \neq \mathsf{NP}$ stands in the way of its own proof?

One could tentatively formalize this creepy idea by the following mathematical statement:

The Terrible Conjecture. If $\mathsf{P} \neq \mathsf{NP}$, then the statement "$\mathsf{P} \neq \mathsf{NP}$" is independent of ZFC.

A proof of the above statement would convince essentially anyone that no-one will ever find a proof that $\mathsf{P} \neq \mathsf{NP}$. However, we do not know how one might prove this terrible conjecture.

But, very surprisingly, one can prove something with a similar flavor!

The natural proofs barrier. If (sufficiently-secure) pseudorandom generators (or one-way functions, or private-key crypto) exist, then the statement "$\mathsf{P} \neq \mathsf{NP}$" cannot be proven by a *natural proof*.

So we have a statement in the same flavor, with a stronger premise, and a weaker/incomparable conclusion.

What is a natural proof?

Let us think of what happens when we prove a lower-bound against a combinatorial model.

We've proven that some function f needs circuits/formulas/whatever of a given size. The proof might go more or less like this:

> We take the function f
>
> bla bla
>
> f has some property A, hence ...
>
> bla bla
>
> and f also has property B, which allows us to conclude ...
>
> bla bla bla
>
> hence f does not have polynomial size circuits.

Then we can consider the predicate $\Phi(f) = A(f) \wedge B(f) \wedge \ldots$, and the proof should work for any f that has the same properties, i.e., such that $\Phi(f) = 1$.

A natural proof is a lower-bound proof such that Φ is a "natural property".

Definition (natural property). **Definition.** Let $S : \mathbb{N} \to \mathbb{N}$, $S(n) \geq n$. An *S-natural property* is a family of predicates $\Phi_n : \{0,1\}^{2^n} \to \{0,1\}$ with three meta-properties:

- (Usefulness) For every $n \in \mathbb{N}$, if a function $f : \{0,1\}^n \to \{0,1\}$ has formulas of size $\leq S(n)$, then $\Phi(f) = 0$. In other words, $\Phi_n(f) = 1$ implies $\mathsf{FSIZE}(f) > S(n)$.
- (Largeness) If we choose a function $f : \{0,1\}^n \to \{0,1\}$ uniformly at random, then $\Pr[\Phi_n(f) = 1] \geq 2^{-O(n)}$. I.e. the property holds for a random function with non-negligible probability. (Notice: non-negligible in the size of f, which is 2^n)
- (Constructivity) The property Φ is decidable in time polynomial in the size of f, i.e., in time $2^{O(n)}$.

Informal definition (natural proof). If:

- We have a lower-bound proof, showing that a family of functions $f_n : \{0,1\}^n \to \{0,1\}$ has $\mathsf{SIZE}(f_n) > S(n)$,
- and from this proof we can somehow *extract* an S-natural property Φ,
- and this proof can be adapted to show that $\Phi(f) = 1$

then we call this proof a natural proof.

The natural proofs barrier (Razborov and Rudich, 1997). If pseudorandom generators $G : \{0,1\}^n \to \{0,1\}^{2n}$ exist having 2^{n^ε} security, for some $\varepsilon > 0$, then no natural proof can show that any function needs super-polynomial size circuits. In fact, under the assumption, it follows that there do not exist natural properties.

I.e., under widely believed assumptions, there exists no natural proof that $SAT\mathsf{NP} \not\subseteq \mathsf{P/poly}$ SAT

Proof of the barrier. It can be shown that one can take a pseudorandom generator G with stretch $2n$ and security 2^{n^ε}, and use it to obtain a pseudorandom family of functions $f_s : \{0,1\}^n \to \{0,1\}$, where $|s| = \mathsf{poly}(n)$, such that, for every algorithm D running in time $2^{O(n)}$,

$$|\Pr_s[D(f_s) = 1] - \Pr_f[D(f) = 1]| < 2^{cn}$$

(one can make c as large as we want by increasing the exponent in the size of s).

This is shown using the same argument of Goldreich-Goldwasser-Micali that we have seen. In that construction, the distinguisher only has oracle access to f. Now we allow the distinguisher to see the entire function. For the GGM pseudorandom function to still be secure against these stronger distinguishers, we use the greater security of the original generator G. NOTE THAT: every function f_s in the pseudorandom family is easy, in particular, it can be computed by a circuit of size polynomial in n.

Now suppose, for contradiction, that there exists a natural property Φ. From usefulness, we conclude that $\Phi(f_s) = 0$ always, because every f_s is easy. From largeness, we conclude that $\Phi(f) = 1$ with non-negligible probability, for a random f. From constructivity, we know that Φ can be computed by a circuit of size $2^{O(n)}$. Ah, but then the circuit that computes Φ can be used as a distinguisher!

That's the proof. In a short one-paragraph explanation, it would be this:

> If we have sufficiently strong PRGs, we can construct a family of functions f_s, all of these are easy, but they look hard to any efficient distinguisher.

Carimo. Ok, so what? Maybe we'll prove it via a "non-natural proof" or whatever.

Remarkably, unbelievably, outrageously, practically every lower-bound proof that we know *is natural*. From experience I can say: it is not at all obvious how to circumvent the natural proofs barrier.

Let us see two examples that we say in class :

- Parity needs formulas of size $n^{3/2}$
- Andreev's function needs formulas of size $n^{5/2}$

Do you remember which property of parity was used to show that it needs such large formulas?

What else? Things we have not seen

- Various complexity classes around non-determinism:
  - The polynomial hierarchy Σ_i^p, Π_i^p, S^2, ...
  - Counting classes: #P, GapP
  - The characterization of NP-search problems PPAD, PPP, CLS, EOPL, etc. These classes allow us to exactly pinpoint the complexity of computational search problems, e.g. I give you the description of a game, and I ask you to compute the Nash equilibrium. This problem is PPAD-complete. Certain problems around lattice-based crypto are PPP-complete. Etc etc...
- Various generalizations of the notion of mathematical proof, which can only be understood, really, with knowledge of computational complexity:
  - Probabilistically checkable proofs (PCPs)
  - Merlin-Arthur and Arthur-Merlin protocols
  - Interactive Proofs
  - Zero Knowledge proofs
- Other combinatorial models of complexity:
  - Constant-depth Boolean circuits
  - Threshold Circuits
  - Decision tree complexity
  - Branching program size
  - Communication Complexity
  - Polynomial degree, approximate degree, threshold degree
  - Discrepancy, Corruption bounds, the γ_2 norms
  - Streaming algorithms
  - Property testing
- Quantum computing: quantum circuits, quantum communication complexity, quantum decision trees, ...
- Algebraic complexity: algebraic formulas and circuits, and lower-bounds for them
- Lots of nice combinatorics that is used:
  - The probabilistic method of Erdos

- Expander graphs
- Isoperimetric inequalities in the Boolean cube (!)
- Randomness extraction, Randomness condensing
- Fourier analysis of Boolean functions
- Compression: source compression, lempel-ziv, ...
- Error correction: error correcting codes, etc

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- (Largeness) If we choose a function $f : \{0,1\}^n \to \{0,1\}$ uniformly at random, then $\Pr[\Phi_n(f) = 1] \geq 2^{-O(n)}$. I.e. the property holds for a random function with non-negligible probability. (Notice: non-negligible in the size of f, which is 2^n)
- (NP-Constructivity) The property Φ is decidable in ***non-deterministic*** time polynomial in the size of f, i.e., in time $2^{O(n)}$.

Rudich's conjecture. There exists a polytime-computable PRG $G : \{0,1\}^n \to \{0,1\}^{2n}$ which can fool NP distinguishers, meaning, there exists $\varepsilon > 0$, such that every non-deterministic algorithm running in time 2^{n^ε},

$$\Pr[A(G(x)) = 1] - \Pr[A(y) = 1] < 2^{-n^\varepsilon}$$

...